# Khyati Khaitan

AI Product Engineer | Generative AI, Agentic AI & Conversational / Voice AI

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## SUMMARY

AI Product Engineer with 4 years of software engineering experience, specializing in production-grade Generative AI, agentic AI workflows (LangGraph, Google Gemini), RAG, and real-time conversational and voice AI systems. Experienced in taking LLM-powered features from prototype to live multi-tenant deployments with FastAPI, Redis, and MongoDB.

## EXPERIENCE

**AI Product Engineer** | VoicePlug Inc., Remote — December 2025 – Present

- Develop and deploy production-grade AI voice ordering features for a multi-tenant QSR platform across multiple live restaurant deployments.
- Increased live ordering success rates from 25–30% to over 60% by resolving entity ambiguity, menu-understanding failures, ASR inconsistencies, and order-validation edge cases in POS-integrated systems.
- Engineered and optimized scalable real-time ordering workflows using FastAPI, Socket.IO, Redis, MongoDB, and Jinja2-based prompt templating, improving intent recognition, fallback handling, and conversational flow reliability.
- Analyzed 500+ real customer interactions with product, QA, and support teams, translating insights into production improvements that reduced ordering friction.

**Software Engineer** | Bank of America, Gandhinagar, Gujarat — July 2022 – November 2025

- Developed a Rasa-based real-time backend chatbot for team member access retrieval, achieving a 100% reduction in manual checks and earning senior management recognition for time and effort savings.
- Created a database parameter comparison utility that eliminated 100% of mismatches across environments, streamlining analysis and ensuring data consistency.
- Led a high-impact domain decommissioning project, performing recursive dependency analysis on 140+ files and refactoring 50+ legacy Python scripts, reducing infrastructure maintenance costs by 20%.
- Conducted detailed regression analysis and ORA-error analysis post-patch, preparing comprehensive reports to validate system integrity and successful patch application.

**Software Intern** | Oracle Cerner, Bengaluru, Karnataka — August 2021 – April 2022

- Contributed to the Data Migration team, extracting, transforming, and exporting confidential clinical data from the Millennium application to client systems.
- Authored and modified HL7 and CCL scripts to meet data migration requirements for diverse datasets.
- Utilized Unix shell scripting with awk, grep, and sed to process and store data extracted from Millennium systems.

## PROJECTS

**Finance News Research Tool (RAG)** — 2025

*Python, LangChain, FAISS, Hugging Face, Groq (GPT-OSS 20B), Streamlit* | github.com/khyatikhaitan/Finance-News-Research-Tool

- Built a RAG-based finance news Q&A tool that extracts article text from URLs and splits it into overlapping chunks.
- Generated Hugging Face MiniLM embeddings and indexed chunks in FAISS for persistent semantic search.
- Served source-grounded answers via GPT-OSS 20B on Groq, showing retrieved source URLs in Streamlit.
- Added multi-turn chat with follow-up questions and source citations for each answer.

**Bengali-to-Hindi Text Refiner** — 2025

*Python, PyTesseract, M2M100, Flan-T5*

- Designed a pipeline to extract handwritten Bengali text from images and translate it into fluent, readable Hindi.
- Used PyTesseract for OCR, M2M100 for multilingual translation, and Flan-T5 to improve fluency and readability.

## SKILLS

**Languages:** Python, Java, SQL, Shell Scripting

**AI, GenAI & NLP:** Generative AI, LLMs, Agentic AI, RAG, LangChain, LangGraph, Google Gemini, Groq, Hugging Face, FAISS, Prompt Engineering, Conversational AI, Voice AI

**Frameworks & APIs:** FastAPI, Socket.IO, Streamlit, Django, Django REST Framework, Rasa

**Databases & Caching:** MongoDB, Redis

**Tools & Platforms:** Git, Bitbucket, Jira, Linux/Unix, VS Code, PyCharm, Google Colab

**Concepts:** Semantic Search, Vector Embeddings, Text Chunking, Intent Recognition, Entity Extraction, ASR Error Handling, Fallback Handling, Prompt Templating, POS Integrations, Real-Time Voice AI Systems

**Problem Solving:** Data Structures & Algorithms – 570+ problems solved on LeetCode

## EDUCATION

**Master of Computer Applications** – Vellore Institute of Technology | CGPA: 9.13 — August 2020 – July 2022

**Bachelor of Computer Applications** – West Bengal University of Technology | CGPA: 8.2 — August 2016 – July 2019